NeuroSleep — Product Concept & Design Direction

A concept-stage design direction, not a finished engineering specification.

Honest note on stage: *no clinical validation, no regulatory submission, and no diagnostic capability exist yet. These would be later milestones pursued with proper testing and the right partner — not claims to make at the concept stage.*

1. Sensing

Pressure mapping across the pillow surface to detect sleep position and shifts in posture, plus an optical heart-rate sensor (PPG) for basic physiological signal. This combination is chosen because it's plausible in a soft, passive-feeling pillow form factor — unlike scalp-contact sensing (e.g. EEG), which realistically doesn't work well from a pillow surface and isn't part of this concept.

2. Adaptive Mechanics

- Adjustable internal support structure to change height/firmness in response to detected posture.
- Active thermal element for temperature regulation through the night.
- On-device control loop connecting sensor input to actuator response — the core technical challenge, and the reason a hardware/embedded-systems partner is essential.

3. Companion App (Design Goal)

Sleep quality reports, manual override for support/firmness/temperature, and personalized recommendations based on the user's own accumulated data. This is a design goal for a later build stage, not a built product.

4. What's Deliberately Not Specified Yet

Exact sensor part numbers, actuator mechanisms, battery/power design, and manufacturing materials are all open — these are exactly the decisions a hardware partner would lead, and specifying them prematurely without that expertise would be guesswork presented as engineering.

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Concept-stage project. No prototype, patent, or team beyond the founder exists yet. Market figures cited from named third-party sources.